MILITARY SPECIFICATION

MOTOR FIRE CONNECTOR ASSEMBLY

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification covers the requirements for Sparrow Motor Fire Connector Assembly, BuWeps LD No. 614000.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effection date of invitation for bids or request for proposals, form a part of this specification.

SPECIFICATIONS

Military

MIL-C-5015	Connectors, Electric "AN" Type
MIL-E-17555	Electrical Equipment Preparation for Delivery of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 1338

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MIL-STD-202 — Test Methods for Electronic and Electrical Component Parts

DRAWING

Bureau of Naval Weapons, Department of the Navy

LD 614000 — Motor Fire Connector Assembly (For Sparrow Rocket Motor)

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government Inspector. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code 105), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General Requirements. -The Motor Fire Connector Assembly shall meet all the requirements of this specification and the requirements specified in the drawings, specifications and other documents listed in LD No. 614000.

3.1.1 Materials. -Materials shall conform to applicable specifications as listed on LD 614000. When drawings permit the use of equivalent material or items, authorization for a specific alternate shall be obtained from the procuring activity.

3.1.2 Finish. -All components shall be free of burrs and contamination which could result in malfunction of the unit or be a safety hazard in handling.

3.1.3 Dimensions. -Unless otherwise specified, dimensions and tolerances are in inches, and apply to the end product after manufacture, thermal treatments, protective coatings, plating, and testing.

3.1.4 Manufacturing Processes. -Manufacturing processes used to produce Motor Fire Connector Assemblies shall be capable of uniformly meeting all requirements of this specification and documents listed in LD No. 614000. Any process change after approval of the preproduction sample shall be reported to and be approved by the procuring activity prior to incorporating the change into manufacture (see 6.2).

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3.2 Preproduction Sample. -Unless otherwise specified in the contract or order, manufacturer who has not produced the Motor Fire Connector Assembly drawing 2435661 previously, shall submit a preproduction sample of ten Fire Connector Assemblies and ten each of the single contact plug connectors and ten each of double contact plug connectors used on the Connector and Cable Assembly drawing 2435669. Each connector shall be provided with a 5.0 ± 0.5 inch lead of cable. The sample and components shall have been manufactured using the methods and procedures to be used for the manufacture of production lots. The sample shall be inspected and tested in accordance with Section 4 of this specification for the purpose of determining that the manufacturer is capable of producing reliable units.

3.2.1 Preproduction Sample Requirements. -The fire connectors and their components shall meet the requirements of 3.1 and subparagraphs thereof.

3.2.2 Electrical Requirements. -In addition to electrical requirements specified on the applicable drawings, the preproduction sample fire connectors and components shall meet the electrical requirements of the following subparagraphs.

3.2.2.1 Dielectric Strength. -The fire connectors and components shall withstand 1000 volts (AC, 60 cps) for 10 seconds minimum between the pin and plug in connector plug, drawing 2478021, and between the two pins in the dual pin plug of drawing 2435669.

3.2.2.2 Surge Current. -The fire connectors and components shall withstand 100 minimum applications of surge current of 20 volts DC at density of 23.0 ± .02 amperes. Each application shall have a duration of .5 seconds minimum. The time between the current applications shall be 15 seconds maximum.

3.2.3 Actuator Assembly. -After assembly of Motor Fire Connector Assembly in the Sparrow rocket motor, the Actuator Assembly drawing 2435662 and connecting components shall move freely without binding.

3.2.4 Thermal Shock. -Each Connector and Cable Assembly of the Motor Fire Connector Assemblies shall meet the drawing and specification requirements after being subjected to the Thermal Shock test specified in 4.4.2.1.

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3.2.5 Vibration. -The individual connectors as specified in 3.2 shall meet the drawing and specification requirements after the vibration test specified in 4.4.2.2.

3.2.6 Physical Shock. -Individual connectors as specified in 3.2 shall meet the drawing and specification requirements after the physical shock test specified in 4.4.2.3. The potential drop at each contact of the mated connector shall be 12 millivolts maximum. The test current shall be 1.00 ± .05 amperes, DC.

3.2.7 Renewal of Preproduction Sample. -In the event that production has been interrupted for more than six months or if there have been major changes which in the opinion of the procuring agency may adversely affect the characteristics of the fire connector, an additional preproduction sample may be required.

3.3 Workmanship. -All workmanship shall meet the minimum quality standards necessary to assure that Motor Fire Connector Assemblies conform to the requirements of this specification and the documents listed in LD No. 614000. The workmanship exhibited in the preproduction samples will be evaluated to determine product acceptability. Approved standards of workmanship will become minimum workmanship requirements for all subsequent units offered by the contractor for acceptance.

3.4 Data Requirements. -No data is required by this specification (other than reports accompanying samples submitted for qualification testing) or by applicable documents referenced in section 2 unless specified in the contract or order (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified the supplier is responsible for the performance of all inspection requirements prior to submission of product for Government inspection and acceptance. Unless otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be complete and be made available to the Government as specified in the contract or order. The Government reserves the right to perform any inspection deemed necessary to assure conformance of supplies and services to prescribed requirements.

4.2 Sampling. -Unless otherwise specified, and when applicable, the sampling plans and procedures used in determining acceptability of products shall

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be in accordance with the provisions set forth in MIL-STD-105.

4.3 Lots. -Lot definition, formation and size shall be in accordance with MIL-STD-105.

4.4 Preproduction Sample Inspection and Tests. -The preproduction sample specified in 3.2 shall be forwarded to the activity designated by the procuring activity for inspection and tests. Further production of the Motor Fire Connector Assembly, LD 614000, by the contractor prior to the completion of the preproduction inspection and tests or to approval by the Bureau of Naval Weapons shall be at the manufacturer's risk. Acceptable Motor Fire Connector preproduction units shall not be counted as part of the quantity specified in the contract or order.

4.4.1 Preproduction Inspection. -The preproduction sample assemblies and components shall be inspected to requirements of 3.1.

4.4.2 Preproduction Tests. -The following tests shall be performed on all preproduction units and individual plug connectors as applicable.

4.4.2.1 Thermal Cycling Test. -Fire Connector Assemblies drawing 2435661 shall be exposed to five (5) continuous cycles of temperature changes in accordance with Method 102 Condition C of MIL-STD-202, except that the temperature range shall be -65°F to 160°F.

4.4.2.2 Vibration Tests. -Each connector plug, single contact, drawing 2478021, and the double contact plug of drawing 2435669 shall be mated with receptacles provided by the procuring activity. Each plug and receptacle coupling shall be tested in accordance with test method 1 of MIL-C-5015.

4.4.2.3 Physical Shock Test. -Each plug and receptacle coupling shall be tested in accordance with the physical shock test specified in MIL-C-5015. The contact potential drop shall be as specified in 3.2.7.

4.4.2.4 Electrical Tests. -All the fire connectors and components shall be tested for requirements of 3.2.2 and subparagraphs thereof. The testing shall be done after the assemblies and parts have passed the tests of 4.4.2.1, 4.4.2.2, and 4.4.2.3.

4.4.2.5 Actuator Test. -The fire connector assemblies shall be assembled to a test fixture simulating attachment to the rocket motor as shown in 2435661. The connector assembly shall meet the requirements of 2435661. The test fixture shall

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meet the dimensional requirements of 2435639 applicable to installation of the fire connector assembly.

4.4.3 Preproduction Acceptance. -Failure of any Motor Fire Connector Assembly, subassembly or part to meet any requirement may result in rejection of the sample and disqualification of the contractor. Final acceptance of the preproduction sample shall be made by the Chief of the Bureau of Naval Weapons.

4.5 Production Inspection and Tests. -The contractor shall be responsible for meeting all the requirements as specified in 3.1. The Classification of Defects (CD) 2435661 shall be the minimum acceptable inspection to be performed. Assemblies shall be serialized and inspection data shall be traceable to a specific serial number (see 6.2).

4.5.1 Production Acceptance. -Based on MIL-STD-105, the Acceptable Quality Level (AQL) for the CD requirements shall be as follows:

(a) Major: 100 percent

(b) Minor: AQL 2.50 percent

5. PREPARATION FOR DELIVERY

5.1 Preservation, Packaging and Packing. -Preservation, packaging, and packing shall be level A.

5.1.1 Level A. -Motor Fire Connector Assemblies shall be packaged and packed in accordance with MIL-E-17555.

5.2 Marking. -Unless otherwise specified by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129 and shall include but not be limited to the following (see 6.2):

(a) Manufacturer's name and address

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(b) Gross and net weights

(c) Lot number and date of manufacture

(d) The number of this specification

(e) Safety precautions or unusual storage requirements

(f) Purchase order or contract number

6. NOTES

6.1 Intended Use. -This specification is intended to be used in procurement of the Motor Fire Connector Assembly which, when installed in the rocket motor provides electrical connection between the Sparrow rocket motor igniter and the missile launcher.

6.2 Ordering Data. -Procurement documents shall specify but not be limited to the following information:

(a) Title, number, and date of this specification

(b) Type and size of container desired

(c) Place of inspection

(d) Minimum lot size, if applicable

(e) Place of delivery

(f) Request for copies of inspection data, if source inspected

(g) Selection of applicable levels of packing and packaging (see 5.1 and 5.2)

(h) Marking requirements (see 5.3)

(i) Sampling if it deviates from MIL-STD-105

(j) Applicable drawings and latest revisions

(k) Data requirements:

(1) Manufacturing process changes (see 3.1.4)

(2) Production inspection and tests (see 4.5)

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6.3 Conflicting Requirements. -Conflicting requirements arising between this specification or any specifications, publications or drawings listed herein, shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

6.4 Request for Deviation. -Requests for deviation from this specification, applicable drawings, specifications, materials or processes shall be forwarded to the procuring activity prior to manufacture of the affected item or items. All deviations shall be limited to the contract under which they were granted.

6.5 Request for Waiver. -Request for waiver from this specification, applicable drawings, specifications, materials or processes shall be forwarded to the procuring activity. A waiver request shall include the following:

(a) Exact nature of defect

(b) CD number and defect classification

(c) Quantity of items involved

6.6 Value Engineering. -All manufacturers of units covered by this specification are encouraged to submit value engineering proposals to the procuring activity in an effort to reduce cost.

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